

Q:1 – Check that whether string ‘abbab’ belongs to given language or not using CYK algorithm. Show full working. (5 + 5 marks)

$S \rightarrow AB$

$A \rightarrow BB \mid a$

$B \rightarrow AB \mid b$

Q:2 – Construct PDA for below language.

$L = \{ a^i b^j c^k \mid i + j = k \text{ and } i, j, k > 0 \}$

Steps:

- 1. Push all a's**
- 2. Push all b's**
- 3. Pop a's and b's against all c's**
- 4. For acceptance, count of a's and b's = count of c's**